Exercise 5.4

(1)(i) $\frac{1}{5}\left[1-\frac{x}{5}+\frac{x^2}{25}-\frac{x^3}{125}+\ldots\right]$ $|x|<5$

(ii) $\frac{2}{9}\left[1-2\left(\frac{4x}{3}\right)+3\left(\frac{4x}{3}\right)^2-4\left(\frac{4x}{3}\right)^3+\ldots\right]$ $|x|<\frac{3}{4}$

(iii) $5^{\left(\frac{2}{3}\right)}\left[1+\frac{2}{15}(\mathrm{x})^2-\frac{1}{225}x^4+\frac{4}{81\times 125}x^6+\ldots\right]$ $x^2<5$

(iv) $2^{\left(-\frac{2}{3}\right)}\left[1-\frac{x}{3}+\frac{5}{36}x^2-\frac{5}{81}x^3+\ldots\right]$ $|x|<2$

(2) $(1001)^{\left(\frac{1}{3}\right)} \approx 10.00333$

(5)(i) $1+5x+\frac{25x^2}{2}+\frac{125x^3}{6}+\frac{625x^4}{24}+\frac{625x^5}{24}+\ldots$

(ii) $1-2x+2x^2-\frac{4x^3}{3}+\frac{2x^4}{3}-\frac{4x^5}{15}+\ldots$

(iii) $1+\frac{1}{2}x+\frac{1}{8}x^2+\frac{1}{48}x^3+\frac{1}{384}x^4+\ldots$

(6)(i) $4x-8x^2+\frac{64x^3}{3}-64x^4+\ldots$ for $|x|<\frac{1}{4}$

(ii) $-2x-\frac{4x^2}{2}-\frac{8x^3}{3}-\frac{16x^4}{4}-\ldots$ for $|x|<\frac{1}{2}$

(iii) $2[3x+\frac{27x^3}{3}+\frac{243x^5}{5}+\frac{2187x^7}{7}+\ldots]$ for $|x|<\frac{1}{3}$

(iv) $-2[2x+\frac{8x^3}{3}+\frac{32x^5}{5}+\frac{128x^7}{7}+\ldots]$ for $|x|<\frac{1}{2}$

(8) $\left(\frac{15}{16}\right)^{\frac{1}{8}}\simeq 0.99196$ (9) $\frac{28}{3}$ (10) $\frac{1}{2}\log_e^{10}$

Exercise 5.5

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
(4)	(4)	(4)	(4)	(2)	(3)	(1)	(3)	(4)	(1)	(1)	(4)	(4)	(2)	(3)
16	17	18	19	20										
(2)	(2)	(3)	(3)	(2)										

Exercise 6.1

(1)(i) $x^2+y^2=81$ (ii) $\frac{x^2}{81}+\frac{y^2}{36}=1$ (2)(i) $y=\pm 2$ (ii) $x=\pm 3$

(3) $x^{2/3}+y^{2/3}=a^{2/3}$ (4) $k=-24,\quad b=-\frac{1}{4}$ (5) $x^2+y^2=16$